

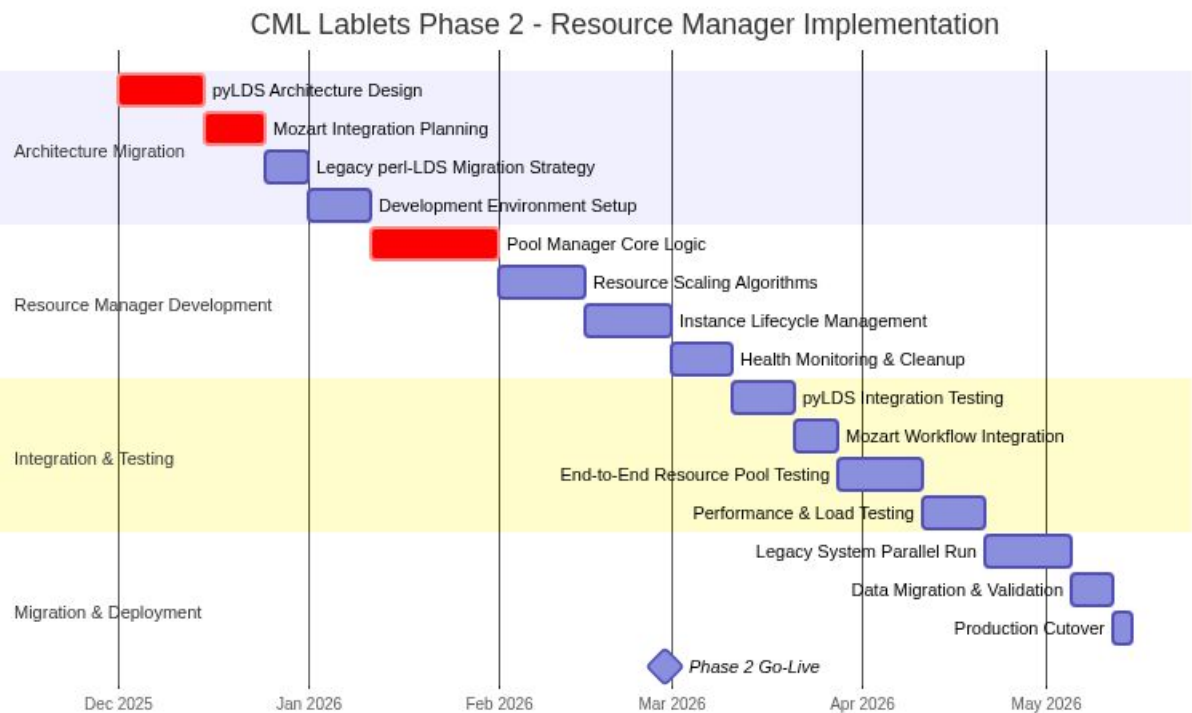
Phase 2: MVP Resource Manager

Timeline: December 2025 - February 2026
Completion Target: FY26Q3 (February 2026)
Project Phase: Resource Management & UX Enhancement

Overview

Phase 2 introduces advanced resource management capabilities to minimize user wait times and improve the overall lab experience. This phase includes migration to modern architecture (pyLDS + Mozart) and implementation of resource pooling to maintain ready-to-use LabInstances.

Detailed Timeline



Key Deliverables

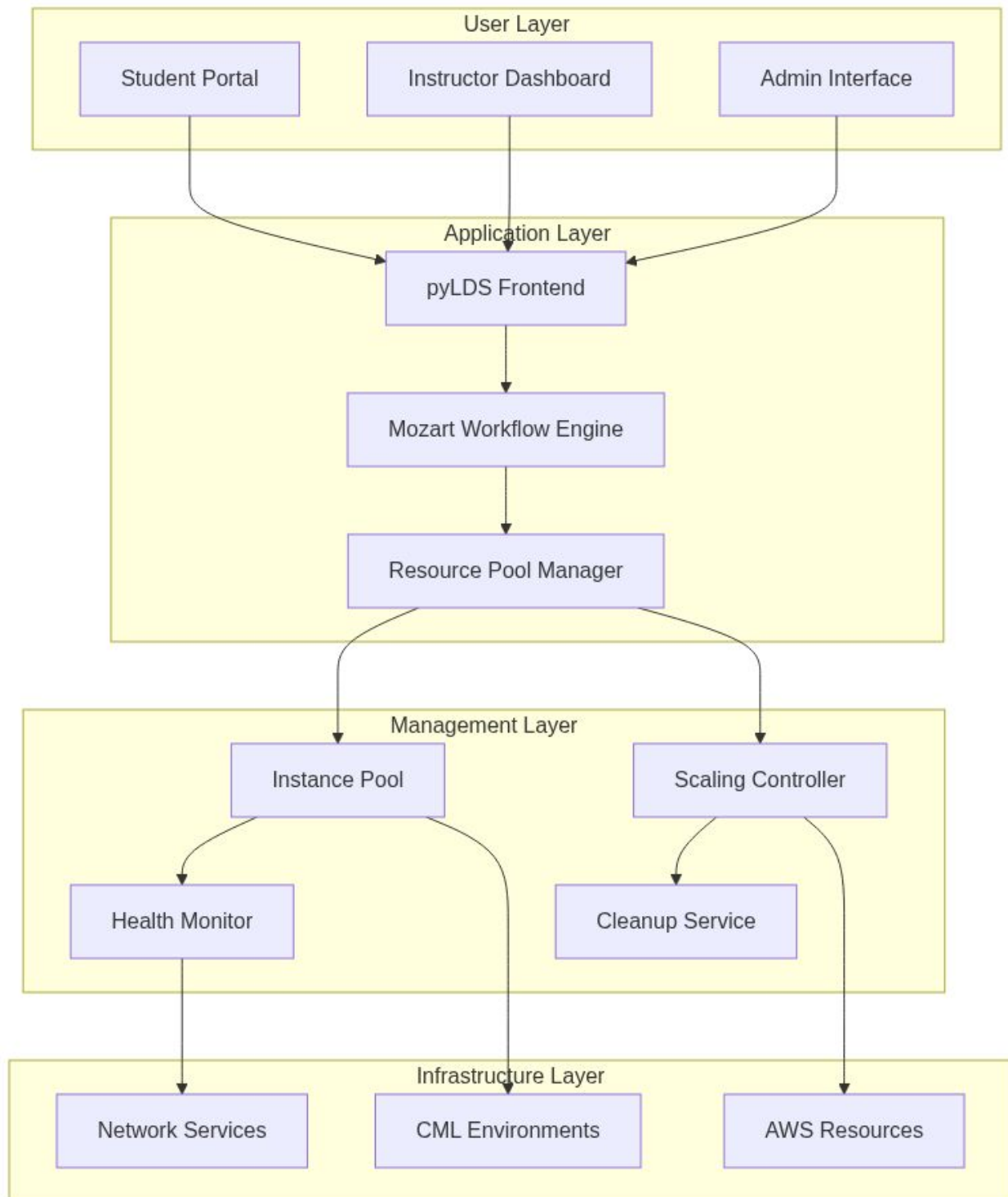
Technical Deliverables

- **Resource Pool Manager:** Intelligent pooling system for LabInstances
- **pyLDS Integration:** Modern learning delivery system implementation
- **Mozart Workflow Engine:** Enhanced automation and orchestration
- **Scaling Algorithms:** Dynamic resource allocation based on demand patterns
- **Monitoring Dashboard:** Real-time visibility into resource utilization

Operational Deliverables

- **Migration Procedures:** Complete cutover from perl-LDS to pyLDS
- **Operational Runbooks:** Enhanced monitoring and troubleshooting guides
- **Performance Baselines:** Established metrics for < 2-minute lab access
- **Support Documentation:** Updated procedures for new architecture

Resource Manager Architecture



Success Criteria & Metrics

Technical Success Criteria

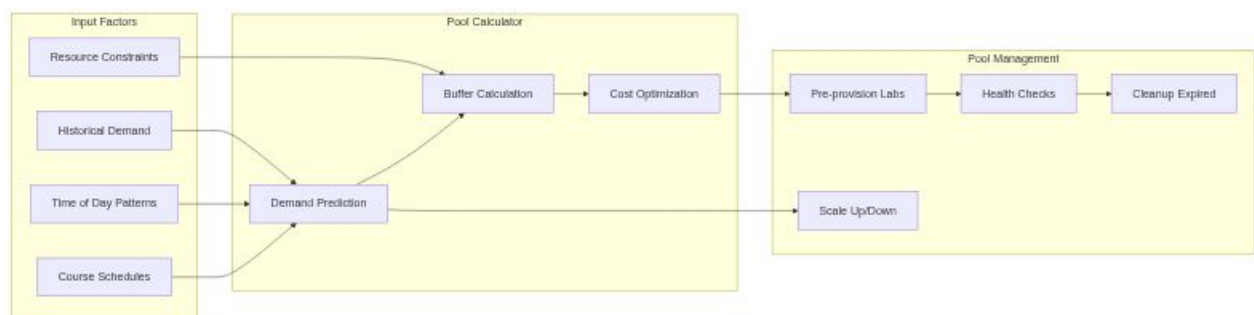
- [] **Lab Access Time:** < 2 minutes from request to ready lab
- [] **System Availability:** > 99.7% during business hours
- [] **Pool Efficiency:** 90%+ of requests served from ready pool
- [] **Resource Utilization:** Optimize costs while maintaining performance
- [] **Migration Success:** Zero data loss during perl-LDS to pyLDS migration

Business Success Criteria

- [] **User Experience:** 95%+ user satisfaction with lab access speed
- [] **Concurrent Users:** Support 20+ simultaneous lab sessions
- [] **Operational Efficiency:** 60% reduction in manual intervention
- [] **Cost Management:** Maintain or reduce infrastructure costs vs Phase 1

Resource Pool Strategy

Pool Sizing Algorithm



Pool Types

1. **Hot Pool:** Fully provisioned and ready labs
2. **Warm Pool:** Pre-configured but not fully initialized
3. **Cold Pool:** Available resources for on-demand provisioning

Migration Strategy

Phase 2A: Parallel Architecture (December 2025)

- Deploy pyLDS alongside existing perl-LDS
- Implement basic resource pooling
- Test integration with Mozart workflows
- Validate performance improvements

Phase 2B: Gradual Migration (January 2026)

- Migrate low-risk lablets to new system
- Monitor performance and user feedback
- Address any integration issues
- Prepare for full cutover

Phase 2C: Complete Migration (February 2026)

- Complete cutover from perl-LDS to pyLDS
- Full resource pool activation
- Remove legacy system dependencies
- Validate all success criteria

Risk Mitigation

Risk Category	Probability	Impact	Mitigation Strategy	Owner
Migration Complexity	Medium	High	Parallel system testing, phased cutover	Technical Lead
Performance Degradation	Low	High	Extensive load testing, rollback plan	Performance Team
Resource Pool Oversizing	Medium	Medium	Adaptive algorithms, cost monitoring	Operations
Integration Issues (Mozart)	Medium	Medium	Early integration testing, API validation	Development
User Adoption Resistance	Low	Medium	Training, change management	Product Team

Key Milestones

timeline
title Phase 2 Milestones
Dec 2025 : Architecture Design Complete
: pyLDS Development Starts
: Resource Pool Design Approved
Jan 2026 : Resource Manager MVP Ready
: Migration Testing Complete
: Parallel System Validation
Feb 2026 : Production Migration Complete
: Performance Targets Achieved
: **PHASE 2 DELIVERY - Feb 28**

Performance Targets

Response Time Metrics

- Lab Request to Ready: < 2 minutes (vs 5 minutes in Phase 1)

- **Pool Replenishment:** < 30 seconds for standard lablets
- **System Response:** < 500ms for API calls
- **Dashboard Load:** < 2 seconds for monitoring interfaces

Resource Efficiency Metrics

- **Pool Hit Rate:** > 90% requests served from ready pool
- **Resource Waste:** < 10% of provisioned resources unused
- **Scaling Response:** < 1 minute to adapt to demand changes
- **Cost Per Session:** 20% reduction vs Phase 1

Technology Stack

Core Components

- **pyLDS:** Modern Python-based learning delivery system
- **Mozart:** Workflow orchestration and automation engine
- **Redis:** Fast caching layer for pool state management
- **PostgreSQL:** Persistent storage for configuration and metrics
- **Prometheus/Grafana:** Monitoring and alerting infrastructure

Integration Points

- **CML API:** Direct integration for lab provisioning
- **AWS APIs:** Resource management and scaling
- **LDAP/SSO:** User authentication and authorization
- **Notification Services:** Email and webhook integrations

Transition to Phase 3

Preparation for Phase 3

- **Multi-tenancy Architecture:** Design for multiple labs per node
- **Advanced Analytics:** Collect detailed usage patterns

- **Cost Optimization Framework:** Establish baseline for Phase 3 improvements
- **Scalability Assessment:** Plan for 100+ concurrent users

Success Handover

- Complete architecture documentation
- Performance benchmarks and optimization opportunities
- User feedback and feature requests for Phase 3
- Operational procedures and lessons learned

Related Documentation

- [Project Timeline Overview](#) - High-level project roadmap
- [Phase 1: MVP APS](#) - Previous phase foundation
- [Phase 3: Cost Optimization](#) - Next phase planning
- [Architecture Overview](#) - Technical architecture evolution